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DATA SCIENCE

ASSIGNMENT - 1



Submitted To,

Prof. Anjana J.

Anjana

Submitted By,

ABHIRAMI. A.S

S3 MCA

Roll no : 02

Date of Announcement : 17/09/22

Date of Submission : 20/09/22

Date of Last Submission : 22/09/22

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1] Given the following data on a certain set of patients seen by a doctor. Can the doctor conclude that a person having chills, fever, mild headache and without running nose has flu? (Use Naive Bayes classification).

Chills	Running Nose	Headache	Fever	Has flu
Y	N	mild	Y	N
Y	Y	no	N	Y
Y	N	strong	Y	Y
N	Y	mild	Y	Y
N	N	no	N	N
N	Y	strong	Y	Y
N	Y	strong	N	N
Y	Y	mild	Y	Y

Consider the above set of certain patients where we can assume set of patients as $P_1, P_2, P_3 \ldots P_8$. We have to find that P_9 patient has flu or not.

P_9 patient has chills, fever, mild headache and no running nose.

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Chills	Running Nose	Headache	Fever	Has flue
Y	N	mild	Y	?

Probability of patients having flue = 5/8

Probability of patients not having flue = 3/8

Calculate probability based on instances.

Chills	Y	N
Y	3/5	1/3
N	2/5	2/3

Running Nose	Y	N
Y	4/5	1/3
N	1/5	2/3

Headache	Y	N
Strong	2/5	1/3
No	1/5	1/3
Mild	2/5	1/3

Fever	Y	N
Y	4/5	1/3
N	1/5	2/3

For finding Pa has flu or not, Calculate,

$$P_{NB}(Y) = P(\text{has flu}=Y) * P(\text{chills}=Y) * P(\text{Runningnose}=N) * P(\text{headache}=\text{mild}) * P(\text{fever}=Y)$$

$$= \frac{5}{8} * \frac{3}{5} * \frac{1}{5} * \frac{2}{5} * \frac{4}{5} = 0.024$$

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$$P_{NB}(N) = P(\text{has flue} = N) * P(\text{chills} = Y) * P(\text{Running nose} = N) * P(\text{Headache} = \text{mild}) * P(\text{fever} = Y)$$

$$= \frac{3}{8} * \frac{1}{3} * \frac{2}{3} * \frac{1}{3} * \frac{1}{3} = \underline{0.009}$$

$$P(Y) = \frac{P_{NB}(Y)}{P_{NB}(Y) + P_{NB}(N)} = \frac{0.024}{0.024 + 0.009} = \underline{0.727}$$

$$P(N) = \frac{P_{NB}(N)}{P_{NB}(Y) + P_{NB}(N)} = \frac{0.009}{0.024 + 0.009} = \underline{0.273}$$

$$P(Y) + P(N) = 0.727 + 0.273 = 1$$

Probability of the patient P9 is resulted as having flu with 0.727 for given instances having chills, fever, mild headache without running nose while comparing to probability of not having flu.

2] Consider a training data set consisting of the fauna of the world. Each unit has 3 features named "Swim", "Fly" and "Crawl". Let the possible values of these features be as follows:

Swim – Fast, Slow, No

Fly – Long, Short, Rarely, No

Crawl – Yes, No

For simplicity, each unit is classified as "Animal", "Bird" or "Fish". Let the training data set be as in the table below. Use Naive Bayes algorithm to classify a particular species if its features are (Slow, Rarely, No).

Sl. No.	Swim	Fly	Crawl	Class
1	Fast	No	No	Fish
2	Fast	No	Yes	Animal
3	Slow	No	No	Animal
4	Fast	No	No	Animal
5	No	Short	No	Bird
6	No	Short	No	Bird
7	No	Rarely	No	Animal
8	Slow	No	Yes	Animal
9	Slow	No	No	Fish
10	Slow	No	Yes	Fish
11	No	Long	No	Bird
12	Fast	No	No	Bird

Consider the training data set of fauna given above. We have to find out the 13th fauna under the given features,

Swim = Slow, Fly = Rarely, Crawl = No.

Probability of class Fish = 3/12

Probability of class Animal = 5/12

Probability of class Bird = 4/12

Probability based on features,

Swim	Fish	Animal	Bird
Fast	1/3	2/5	1/4
Slow	2/3	2/5	0/4
No	0/3	1/5	3/4

Fly	Fish	Animal	Bird
Long	0/3	0/5	1/4
Short	0/3	0/5	2/4
Rarely	0/3	1/5	0/4
No	3/3	4/5	1/4

Crawl	Fish	Animal	Bird
Yes	1/3	2/3	0/4
No	2/3	3/5	4/4

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$$P_{NB}(\text{Fish}) = P_{(\text{class}=\text{fish})} \times P_{(\text{swim}=\text{slow})} \times P_{(\text{fly}=\text{rarely})} \times P_{(\text{crawl}=\text{No})}$$

$$= \frac{3}{12} \times \frac{2}{3} \times 0 \times \frac{2}{3} = \underline{\underline{0}}$$

$$P_{NB}(\text{Animal}) = P_{(\text{class}=\text{Animal})} \times P_{(\text{swim}=\text{slow})} \times P_{(\text{fly}=\text{rarely})} \times P_{(\text{crawl}=\text{No})}$$

$$= \frac{5}{12} \times \frac{2}{5} \times \frac{1}{5} \times \frac{3}{5} = \underline{\underline{0.02}}$$

$$P_{NB}(\text{Bird}) = P_{(\text{class}=\text{Bird})} \times P_{(\text{swim}=\text{slow})} \times P_{(\text{fly}=\text{rarely})} \times P_{(\text{crawl}=\text{No})}$$

$$= \frac{4}{12} \times 0 \times 0 \times \frac{4}{4} = \underline{\underline{0}}$$

To check if the obtained solution is correct or not,

$$P(\text{fish}) = \frac{P_{NB}(\text{fish})}{P_{NB}(\text{fish}) + P_{NB}(\text{Animal}) + P_{NB}(\text{Bird})} = \frac{0}{0+0.2+0} = \underline{\underline{0}}$$

$$P(\text{Animal}) = \frac{P_{NB}(\text{Animal})}{P_{NB}(\text{fish}) + P_{NB}(\text{Animal}) + P_{NB}(\text{Bird})} = \frac{0.02}{0+0.02+0} = \underline{\underline{1}}$$

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$$P(\text{Bird}) = \frac{P_{NB}(\text{Bird})}{P_{NB}(\text{fish}) + P_{NB}(\text{Animal}) + P_{NB}(\text{Bird})} = \frac{0}{0+0.2+0} = 0$$

$$\Rightarrow P(\text{fish}) + P(\text{Animal}) + P(\text{Bird}) = 1$$

∴ The obtained solution is correct.

The probability of 13th fauna is resulted as Animal with 1 for given features – Swim = short, fly- rarely, Crawl = No while comparing to other classification.

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